

AR 502: Advanced Design Practicum

Course Code	AR 502
Course Name	Advanced Design Practicum
L-T-P-C	3-0-2-4
Prerequisites	Consent of faculty advisor
Intended for	PG and PhD
Mutual Exclusion	None
Approval	50th BoA (Jul 2023)

Course Contents

- **Introduction:** Engineering design - How to select an engineering problem, stages of solving a problem, documentation in Engineering, Machine and a robot - Different aspects of robotics, current problems in robotics. Drives and motion, pneumatic, hydraulic systems, clutch and brake. (3 hours)
 - Practical 1: Microelectronics, onboard computer, IoT, embedded electronic and materials to be used in the lab
 - Manufacturing techniques - additive and subtractive manufacturing. (3 hours)
- **Design of Mechanical Systems:** Introduction to CAD by 3D modeling software, Drawing of parts and assemblies, Computer-Aided Manufacturing and prototyping, Brief Introduction to robotic systems, Joints and transformations on ROS. (5 hours)
 - Practical 2: Introduction to Mechanical assembly, bill of materials, 3D modeling software and design of parts and assemblies and Static Stress Simulation. (3 hours)
 - Practical 3: Simulation in ROS environment (3 hours)
- **Integration of Intelligent Control:** Sensor and Actuator selection and sizing, Determination of Power Source, Design of Power distribution, microcontroller, and motor driver circuits, Developing PCB boards and feasibility testing, Introduction to standard electronic connectors and American Wire Gauge. (7 hours)
 - Practical 4: Sizing of sensors motors and linear actuators and integration into mechanical design. (3 hours)
 - Practical 5: Circuit Design using CAD tool and making PCBs manually and demonstration of CNC based PCB printing. (3 hours)
- **Programming and Signals:** Introduction to Programming; procedural vs object-oriented programming, Object-Oriented programming in practice, Signals; communication via PWM, UART, Design of hardware and software interrupts. (6 hours)
 - Practical 6: Introduction to Programming; procedural vs object-oriented programming. (3 hours)
 - Practical 7: Signals; communication via PWM, UART; connecting two microcontrollers. (3 hours)
 - Practical 8: Design of hardware and software interrupts. (3 hours)
- **Integration of Compute and networks:** Introduction to microcontroller coding and interfacing with the ros API, Introduction to ROS packages and their deployment, Introduction to IoT and IP sending receiving packets on client-server networks, Control of robotic platforms over IP, Deployment of real-time decision pipelines on the robot. (6 hours)
 - Practical 9: Intro to roserial and connecting microcontrollers to ROS. (3 hours)
 - Practical 10: Introduction to esp8266; control via blynk. (3 hours)
 - Practical 11: Deployment of conditional path planning on robot and testing. (3 hours)
- **Final project:** Project towards design and development of functional Robotic system.

Text Books

1. Owen Bishop, **Robot Builder's Cookbook**
2. Gaurav Verma, **Autodesk Fusion 360**
3. Godfrey C. Onwubolu, **Introduction to SOLIDWORKS: A Comprehensive Guide with Applications in 3D Printing**

References

1. Morgan Quigley, **Brian Gerkey and Programming Robots with ROS**